MIL-S-49301/6(CR)
18 July 1988

MILITARY SPECIFICATION SHEET

SCANNER, MECHANICAL MX-10717(V)6/U

This specification is approved for use by USACECOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification sheet covers the Mechanical Scanner, MX-10717(V)6/U and augments MIL-S-49301. The MX-10717(V)6/U is used on the Bradley Fighting Vehicle, Integrated Sight Unit.

1.2 Purpose. The purpose of the specification sheet is to modify or expand the contents of MIL-S-49301 to describe the MX-10717(V)6/U Mechanical Scanner. All the MIL-S-49301 contents apply except as specifically modified herein. Paragraph numbers directly correlate between MIL-S-49301 and this specification sheet.

2. APPLICABLE DOCUMENTS

2.1 Order of precedence. In the event of a conflict between the text of this specification sheet and the references cited herein, the text of this specification sheet shall take precedence.

SPECIFICATIONS

MILITARY

MIL-S-49301	- Scanner, Mechanical, MX-10717(V)/U, General Specification for

DRAWINGS

USA COMMUNICATIONS-ELECTRONICS COMMAND

80063-SM-D-657598	- Scan-Interlace Assembly, PL-1403A/UA
80063-SM-D-773950	- Scan-Interlace Assembly, PL-1403/UA
54490-5006243-6	- Scanner, Mechanical, MX-10717(V)6/U

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, NJ 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426 appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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3. REQUIREMENTS

3.1 Description. The MX-10717(V)6/U Scanner, herein known as the MX-10717(V)6/U, shall consist of the MIL-S-49301, MX-10717(V)6/U Scanner with a null lens and scanner mount assembled and adjusted for 30 Hz operation and 5.50 ± 0.20 degree total scan angle. The MX-10717(V)6/U scanner can be operated with either scan interlace assembly PL-1403/UA or PL-1403A/UA. These two assemblies will be referred to as the scan interlace assembly throughout this specification sheet.

3.2 Construction. The MX-10717(V)6/U shall be constructed in accordance with drawing 5006243-6.

3.2.1 Weight. Weight of the MX-10717(V)6/U shall be 1.73 pounds maximum.

3.6 Performance characteristics. The characteristics specified herein shall apply as constructed (3.2). Operation shall be at a nominal 30 Hz, 2:1 side interlace mode, while using the Scan-Interlace Assembly.

3.6.1 Scan angle and frequency.

3.6.1.1 Variable scan frequency. Not applicable.

3.6.1.2 Specific scan frequency (30 Hz). Scan frequency of the MX-10717(V)6/U with the Scan-Interlace Assembly shall be adjusted to 30.0 ±1.0 Hz. Scan frequency drift shall not be greater than ± 1.5 Hz from the frequency at ambient setting (see MIL-S-49301, 4.4) when exposed to high and low temperature environmental conditions (MIL-S-49301).

3.6.1.3 Synchronous scan frequency. Not applicable.

3.6.1.4 Scan angle. The scan angle shall be 2.75 ± 0.10 degrees in each direction (5.50 ± 0.20 degrees total) from a true 45-degree position with respect to the mounting surface (Datum -A- in figure 5) required for securing the MX-10717(V)6/U.

3.6.1.5 Scan transducer D.C. offset. The scan position transducer shall have a D.C. offset of 0.0 ±0.2 Vdc.

3.6.2 Interlace. The MX-10717(V)6/U shall operate in the 2:1 side interlace mode. The interlace switching time shown in MIL-S-49301, figure 2, shall not be greater than 4.0 milliseconds (ms), and the average amplitude shall be 1.0 ±0.2 V p-p.

3.6.2.1 Interlace angle. The interlace solenoid stops and scan-interlace assembly, potentiometers shall be used to determine a dynamic object space interlace angle of 155 ± 15 arc seconds.

3.6.2.2 Interlace solenoid position. The distance of the interlace solenoid assembly from the solenoid plungers, herein defined as the air gap, shall be equivalent to a 2.0 ± 0.2 Vdc swing from a mechanical gimbal zero position, when measured at J3 of the Scan-Interlace Assembly.

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3.6.2.3 Elevation boresight. The scan mirror shall not deviate by more than ± 16 arc minutes from the scanner optical axis. (The optical axis is orthogonal to the scanner mounting surface. See figure 1, DATUM -A-.)

3.6.2.4 Interlace transducer D.C. offset. The interlace position transducer shall have a D.C. offset of 0.0 ± 0.2 Vdc.

3.6.3 Scan direction. The MX-10717(V)6/U shall show the waveform shapes and phase relationships in figure 2 while operating in the 2:1 side interlace mode.

3.6.4 Power consumption. Power required to operate the MX-10717(V)6/U shall not exceed 2.2 watts.

3.6.5 Scan efficiency. The MX-10717(V)6/U scan efficiency shall be a minimum of 75 percent.

3.6.6 Scan jitter in azimuth. Jitter time shall be 62.0 microseconds maximum.

3.7 Environmental conditions. See MIL-S-49301 except as follows:

3.7.4 Shock. See MIL-S-49301 and 3.7.4.1.

3.7.4.1 Shock and scan frequency. Scan frequency (3.6.1.2) shall return to normal within 0.2 seconds after the MX-10717(V)6/U is subjected to the operating shock environment.

3.7.5 Vibration. The MX-10717(V)6/U shall not be damaged by vibration over a frequency spectrum at the specified g levels and amplitudes shown in figure 3 for the X axis and as shown in figure 4 of MIL-S-49301 for the Y and Z axis. During specified vibration, the average interlace waveform amplitude shown in MIL-S-49301, figure 2, shall not deviate from the average amplitude without vibration by more than ± 50 percent.

4. QUALITY ASSURANCE PROVISIONS

4.6 Test methods. Tests shall be performed utilizing the scan interlace assembly with the MX-10717(V)6/U.

4.6.1 Scan frequency.

4.6.1.1 Variable scan frequency. Not applicable.

4.6.1.2 Specific scan frequency (30 Hz). The MX-10717(V)6/U shall be connected to the Scan-Interlace Assembly, and mounted to a suitable fixture. Power shall be applied and the output of the tachometer (Scan-Interlace Assembly) and shall be monitored with a calibrated frequency counter. The output of the Scan-Interlace Assembly shall be adjusted to obtain the specific scan frequency. Failure to meet the requirements of 3.6.1.2 shall constitute faiure of this test.

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4.6.1.3 Synchronous scan frequency. Not applicable.

4.6.1.4 Scan angle.

4.6.1.4.1 Test apparatus. The test apparatus shall consist of; a holding fixture utilizing the mounting surface (DATUM -A- in figure 5) required for securing the MX-10717(V)6/U, a projection system capable of projecting a spot or marker into the scanner along the optical axis (figure 5), and an appropriate scale graduated in degrees measured about the scan mirror axis. The zero position of this scale shall be the position where the spot or marker is projected onto the scale when the mirror is at a true 45 degrees with respect to the mounting surface (DATUM -A- figure 5) required for securing the MX-10717(V)6/U.

4.6.1.4.2 Test conduction. The scanner shall be operated and the angular excursions of the projected spot, in each direction, shall be recorded.

4.6.1.4.3 Failure criteria. Failure to meet the requirement of 3.6.1.4 shall constitute failure of this test.

4.6.1.5 Scan transducer D.C. offset. The D.C. offset of the scan position transducer shall be measured at the scan position test point of the test circuit in Figure 4 with the scan mirror in the mechanical zero position. The D.C. offset shall meet the requirements of 3.6.1.5.

4.6.2 Interlace. With the scanner mounted in a suitable fixture, the output of the interlace position transducer shall be monitored with an oscilloscope. The waveform switching times (figure 1 of MIL-S-49301) and amplitude shall be measured to verify that the requirements of 3.6.2 are met. Failure to meet requirements of 3.6.2 shall constitute failure of this test.

4.6.2.1 Interlace angle. Using a autocollimator and an auxilliary mirror to measure along the optical axis (see figure 5), the interlace solenoid stops and Scan-Interlace Assembly shall be adjusted to scan-angle provide a dynamic object space interlace angle of 155 $\pm$ 15 arc seconds. Failure to meet the requirements of 3.6.2.1 shall constitute a failure of this test.

4.6.2.2. Interlace solenoid position. With the scanner mounted to a suitable fixture and the scanner disabled, the interlace solenoid assembly offset from the plunger shall be measured at J3 of the Scan-Interlace Assembly. The offset shall be equivalent to 2.0 $\pm$ 0.2 Vdc and -2.0 $\pm$ 0.2 Vdc from a mechanical gimbal zero position. The interlace position transducer (MT2) is to have a target to transducer distance of 0.025 $\pm$ 0.005 inches and a bridge transducr voltage of 1.25 $\pm$ 0.1 Vdc. The interlace solenoid stops are to be retracted so that the interlace plungers make contact with the interlace solenoid assembly. Failure to meet the requirements of 3.6.2.2 shall constitute a failure of this test.

4.6.2.3 Elevation boresight. The scan mirror shall not deviate by more than $\pm$ 16 arc minutes in elevation from the optical axis (see figure 1). Elevation boresight shall be measured at the approximate center of the azimuth scan travel with interlace not operating. Failure to meet the requirements of 3.6.2.3 shall constitute a failure of this test.

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4.6.2.4 Interlace transducer D.C. offset. The D.C. offset of the interlace position transducer shall be measured at J3 of the Scan Interlace Assembly. With the interlace gimbal in the mechanical zero position, the D.C. offset shall meet the requirements of 3.6.2.4.

4.6.3 Scan direction. With the MX-10717(V)6/U operating in the 2:1 side interlace mode, the outputs of the interlace transducer and the scan tachometer (Scan-Interlace Assembly) are observed with an oscilloscope. The output waveforms shall show the applicable shape and phase relationships shown in figure 2. Failure to meet the requirements of 3.6.3 shall constitute failure of this test.

4.6.4 Power consumption. See MIL-S-49301.

4.6.5 Scan efficiency. The MX-10717(V)6/U shall be operated with the Scan-Interlace Assembly and measurements taken as follows: The period of the scan tachometer is measured by an oscilloscope. The active time between two consecutive scans shall be determined by measuring the time between the 10 percent and the 90 percent points of the scan tachometer waveform. Scan efficiency shall be calculated by dividing the sum of the two consecutive scans by the scan tachometer period and multiplying by 100 percent. Failure to meet the requirements of 3.6.5 shall constitute failure of this test.

4.6.6 Scan jitter in azimuth. The MX-10717(V)6/U shall be tested as specified in MIL-S-49301 with additional circuitry used as shown in figure 4 to test the apparatus. Failure to meet the requirements of 3.6.6 shall constitute failure of this test.

4.7 Environmental tests.

4.7.4 Shock. See 4.7.4.1 and MIL-S-49301.

4.7.4.1 Shock and scan frequency. Failure to meet the requirements of 3.7.4.1 during the operating test shall constitute failure of this test.

4.7.5 Vibration. See figure 3 and MIL-S-49301.

6. NOTES

6.1 Intended use. The MX-10717(V)6/U is intended for use with the Bradley Fighting Vehicle, Integrated Sight Unit.

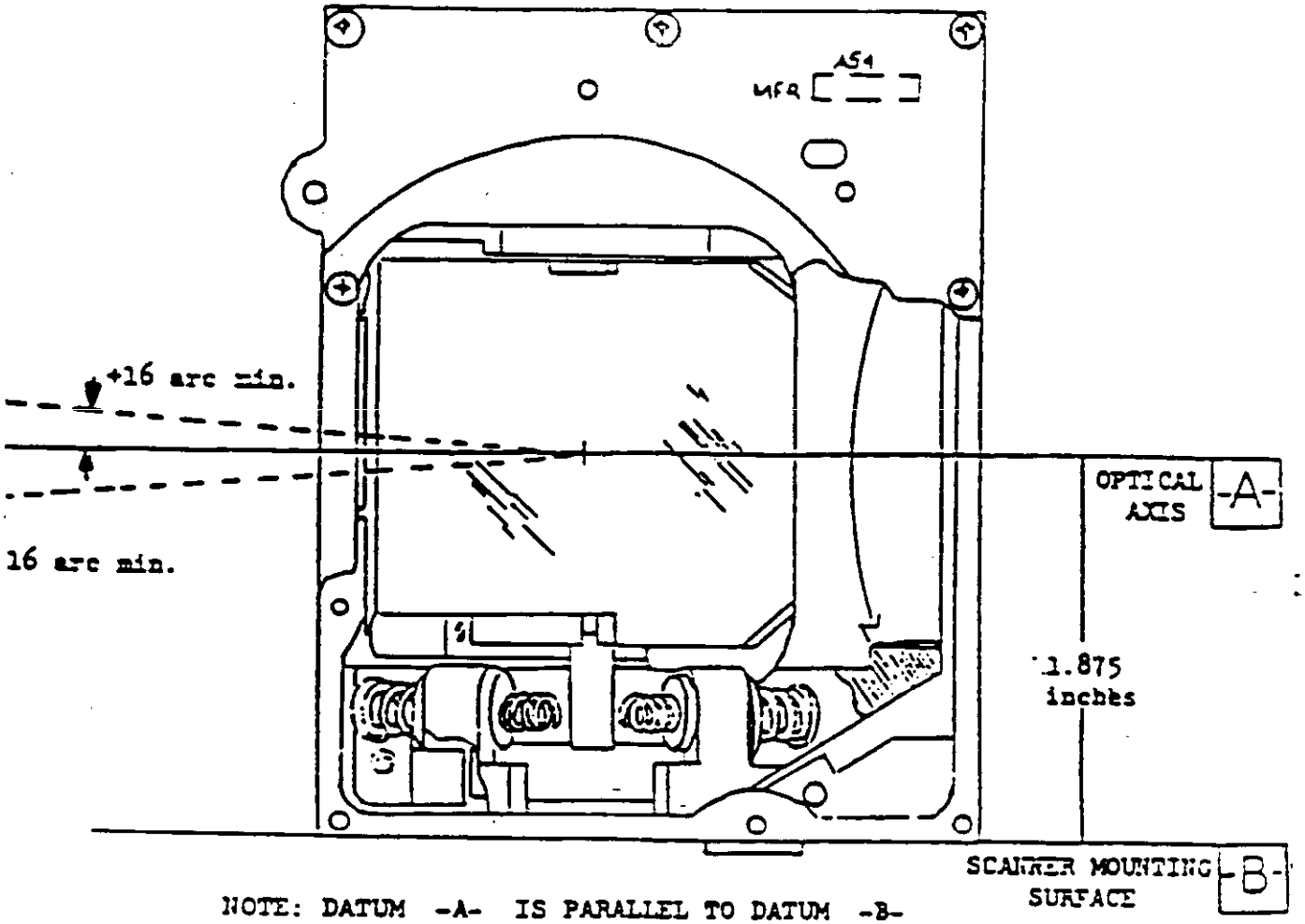
6.2 Interchangeability. The MX-10717(V)6/U scanner built and tested in accordance with this specification and applicable drawings may be used in place of the MX-9872(V)6/UA scanner. The MX-9872(V)6/UA may not be used in place of the MX-10717(V)6/U scanner.

6.3 Subject term (keyword) listing.
FLIR
Sensor
Scanner

CUSTODIAN:
Army - CR

PREPARING ACTIVITY
Army - CR
Project 5855-A290-6

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NOTE: DATUM -A- IS PARALLEL TO DATUM -B-

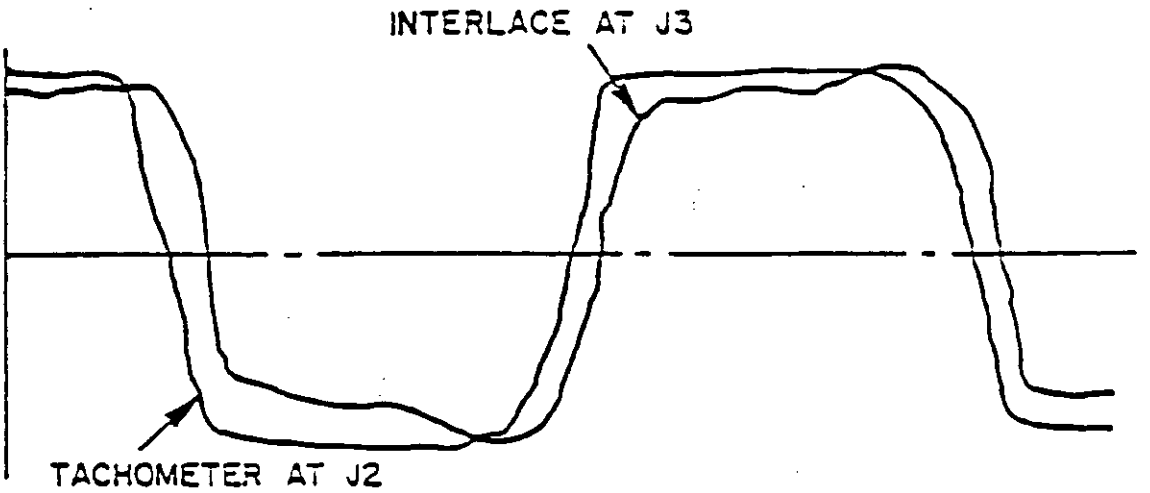
FIGURE 1. ELEVATION BORESIGHT.

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A

2:1 'REAR POSITION'

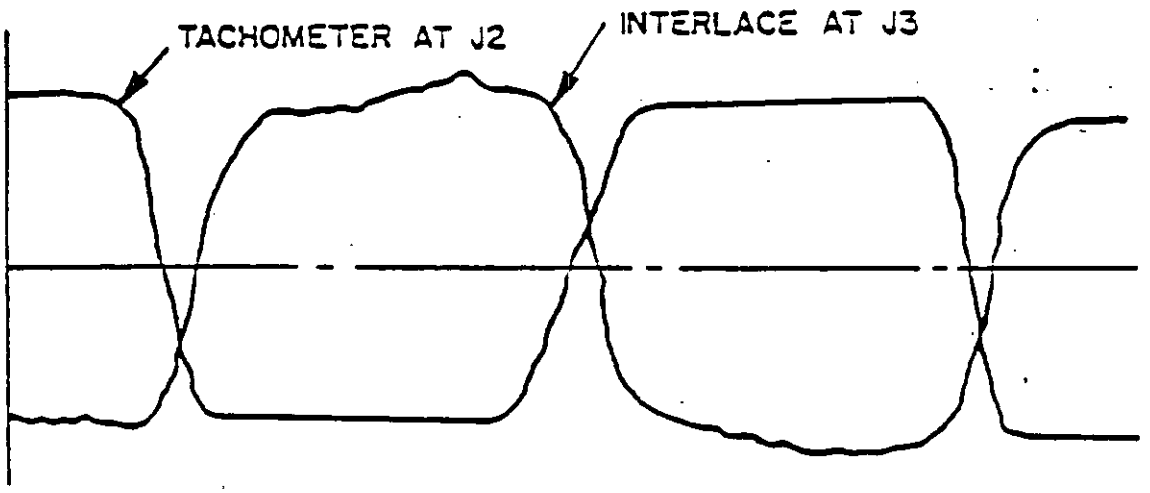
PL-1403/UA
P2: CONNECT
PIN 10 TO 20
PIN 8 TO 18
PIN 17 TO 7
Oscilloscope:
(See Note)



B

2:1 'SIDE POSITION'

PL-1403/UA
P2: CONNECT
PIN 19 to 20
PIN 17 to 18
PIN 8 to 7
Oscilloscope:
(See Note)



NOTE - CONNECT OSCILLOSCOPE AS FOLLOWS:

CHANNEL 1 - (TACHOMETER) HIGH INPUT TO J2 (OR J1-13) AND LOW SCOPE TO GND.
CHANNEL 2 - (INTERLACE) HIGH INPUT TO J3 (OR J1-1) AND LOW SCOPE TO GND.

FIGURE 2. SHAPE AND PHASE RELATIONSHIP OF INTERLACE TRANSDUCER AND SCAN TACHOMETER.

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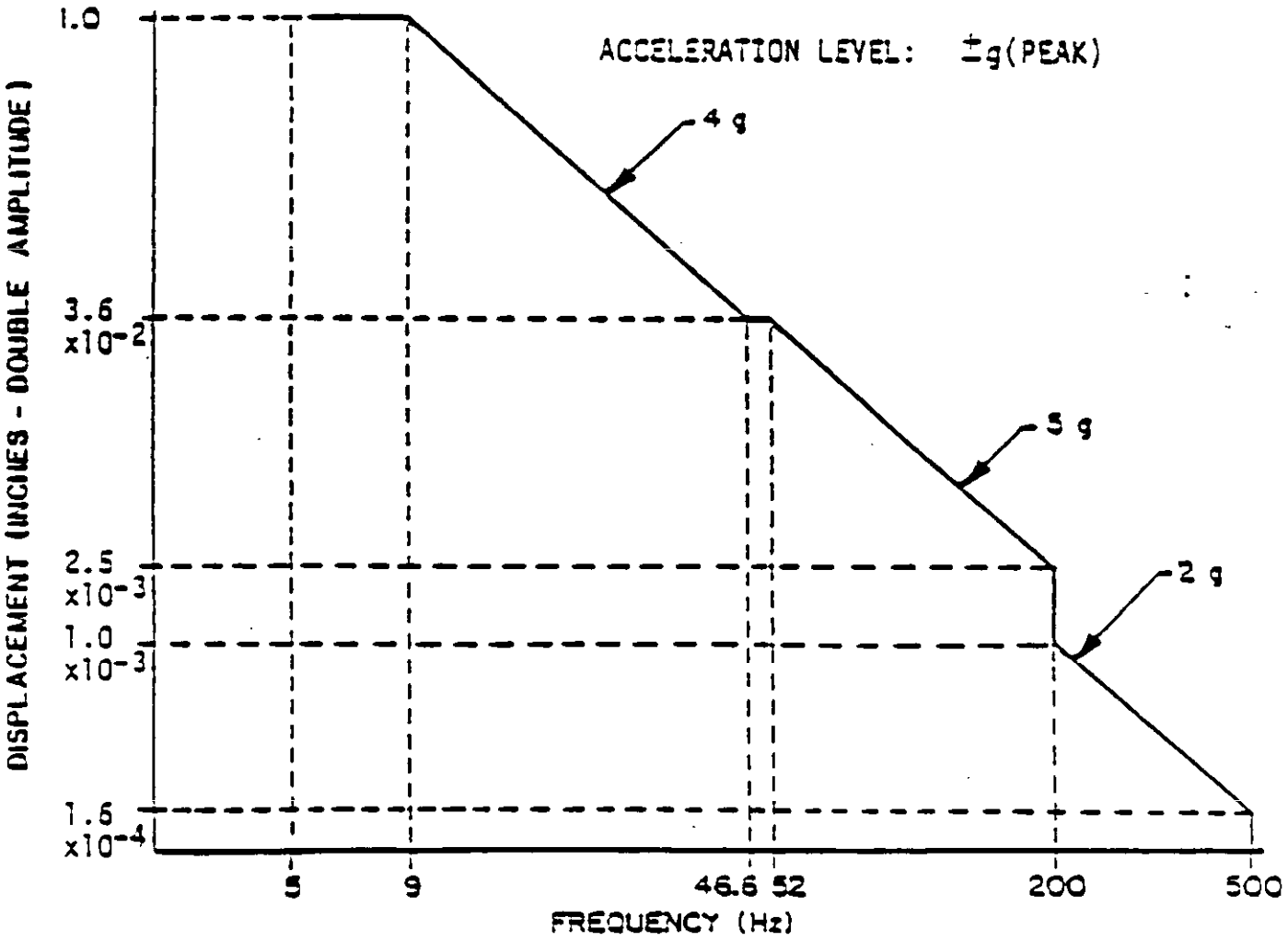


FIGURE 3. VIBRATION TEST PROFILE (X AXIS).

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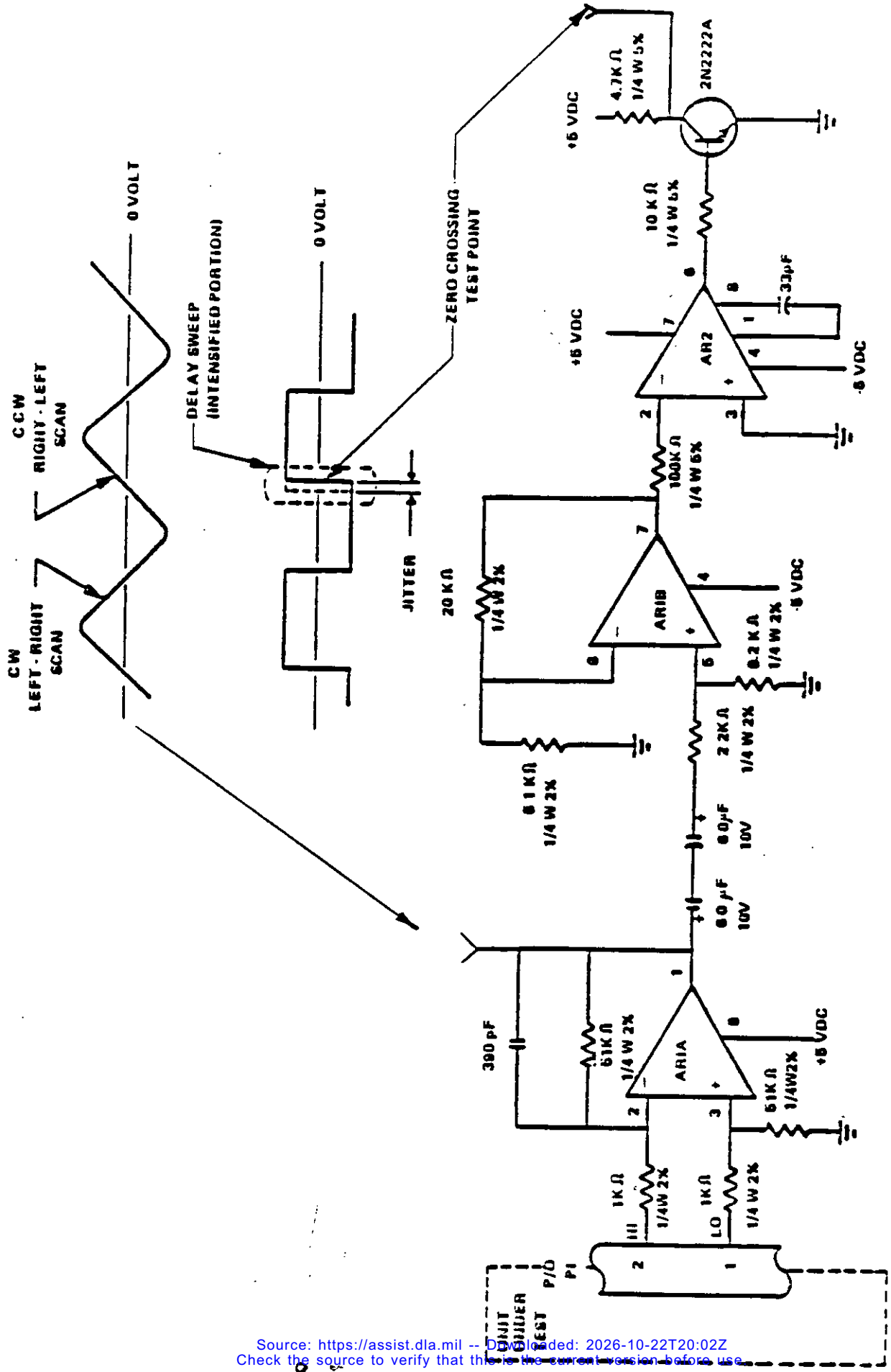
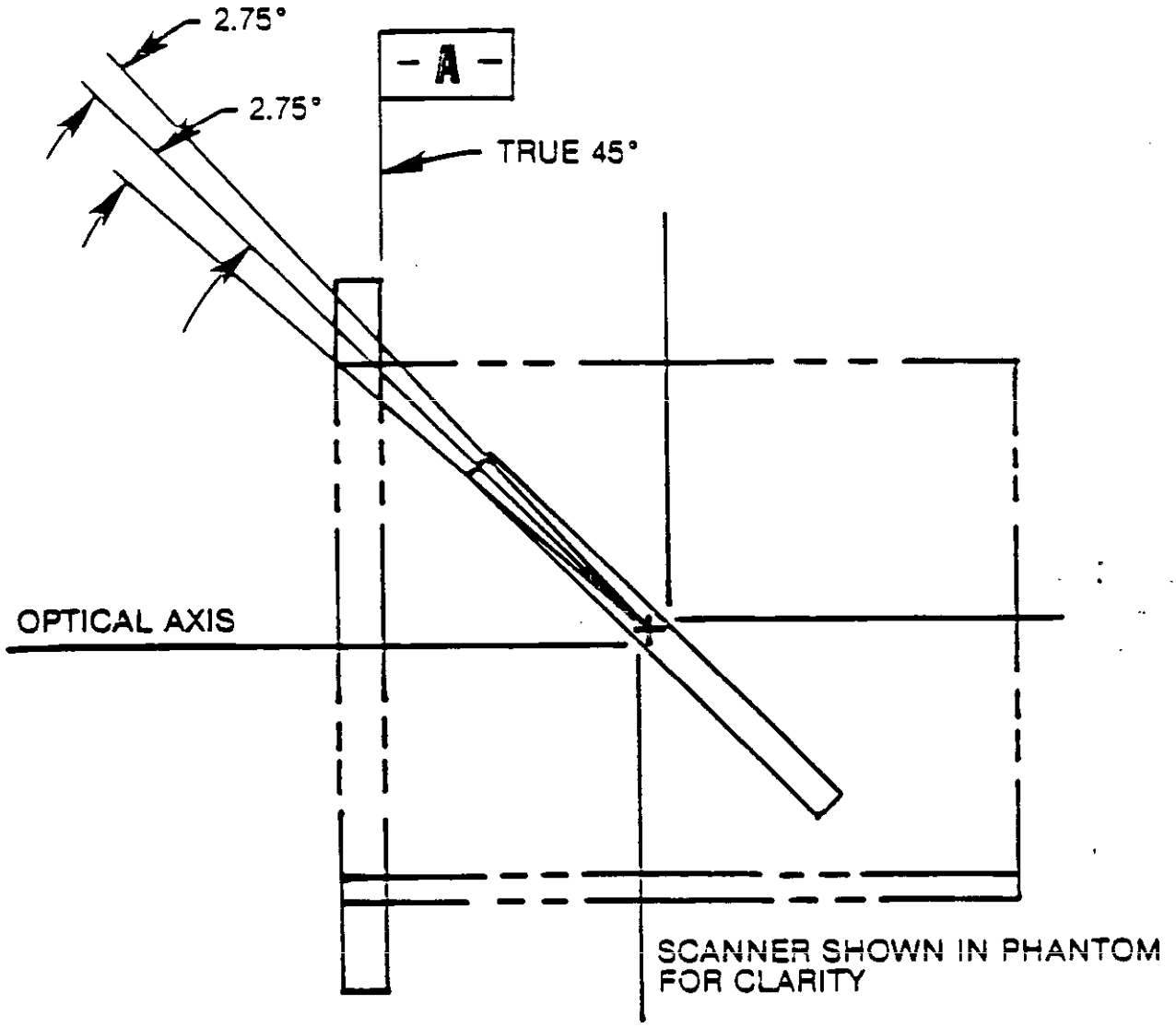


FIGURE 4. TYPICAL CIRCUITRY FOR SCAN JITTER TEST.

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THE OPTICAL AXIS IS DEFINED TO BE ORTHOGONAL TO THE MOUNTING SURFACE, DATUM -A-.

Figure 5. SCAN ANGLE.

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-49301/6(CR)	Scanner, Mechanical MX-10717(V)6/U

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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